Graph Theory

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Overview

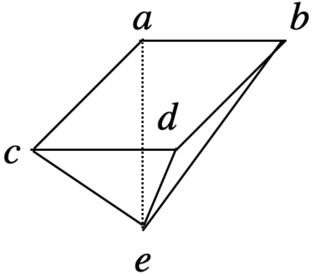
Vertex Adjacency and Degrees (11.1)

Some Common Graphs (11.3)

Graphs and polyhedra

A *graph* consists of a set of vertices (nodes) and edges (links) that connect them.

This terminology comes from descriptions of flat-sided solid shapes—polyhedra.



$\{a, b, c, d, e\}$ are the vertices, $\{\{a, b\}, \{a, c\}, \{c, d\}, \{b, d\}, \{a, e\}, \{b, e\}, \{c, e\}, \{d, e\}\}$ are the edges.

Examples of graphs

A graph can be viewed as a symmetric relation. Examples of edges:

- Two people that have been in direct contact.
- Two intersections connected by a road.
- Two accounts that are Facebook friends.
- Two gates connected by a wire.
- Two translators that speak a shared language.
- Two atoms bonded together.

(There are also directed graphs where the pairs are ordered. But we're not talking about them.)

Graph definition

Definition: A *graph* G consists of a set of vertices $V(G)$ and edges $E(G)$.

$E(G) \subseteq \{\{u,v\} | u,v \in V(G), u \neq v\}$.

u adjacent to v in G iff $\{u,v\} \in E(G)$.

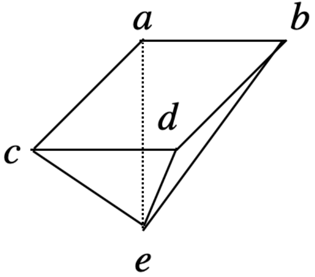
The "adjacent to" relation is reflexive? No. Symmetric? Yes! Transitive? Depends. Empty is graph is! Otherwise, no, because symmetry and non-reflexive.

Degree

Definition: The *degree* of vertex $u \in V(G)$ is

$$\deg(u) = |\{v | \{u, v\} \in E(G)\}|.$$

In words? The number of vertices adjacent to u. The number of edges that include v.

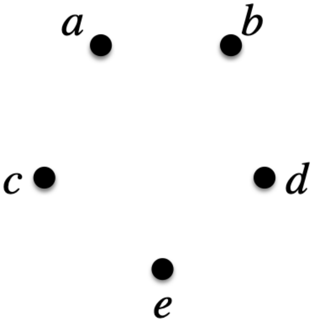


All of the vertices of this graph have the same degree. Yes or no?

Empty graph

Often convenient to let $n = |V(G)|$ and $m = |E(G)|$.

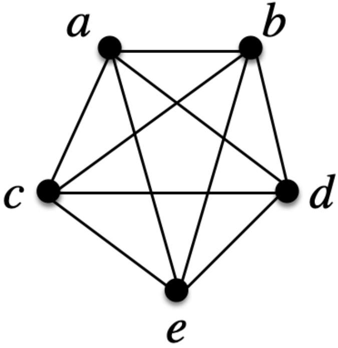
We insist that $n > 0$. But, if $m = 0$, we call it the *empty graph* on n nodes.



Complete graph

If, $\forall u, v \in V(G)$, $\{u, v\} \in E(G)$, we say G is a *complete graph* (or a clique). Also known as K_n.

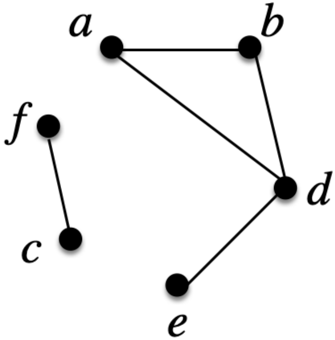
$m = \binom{n}{2} = n(n-1)/2$.



Path

A length-k path p in G is

- an element of $V(G)^{k+1}$
- such that $\forall i \in [1, k], \{p^i, p^{i+1}\} \in E(G)$.
- The path is *simple* if there is no $i \neq i'$ such that $p^i = p^{i'}$.

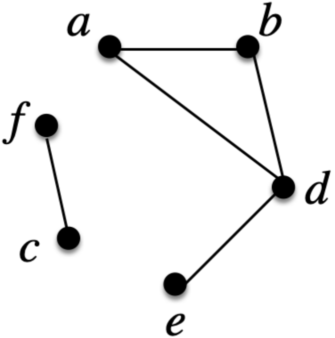


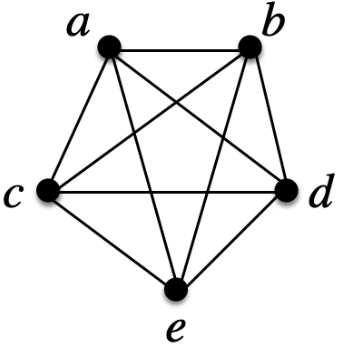
Paths? (a, b, d, a)? Yes (not simple). (b, a, e)? No, missing edge.

Connected graph

A pair of vertices u and v is *connected* if there exists a length-k path such that $p^1 = u$ and $p^{k+1} = v$.

A graph G is connected if all pairs of vertices in $V(G)$ are connected: $\forall(u, v) \in V(G), \exists(u, v)$-path

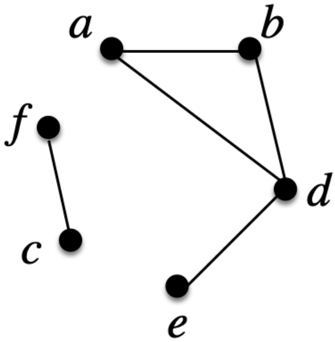




Cycle

A length-k cycle p in G is

- a length-k path in G
- where $p^{k+1} = p^1$.



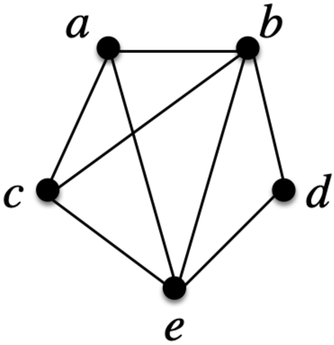
Cycles? (a, b, d, a)? Yes. (b, a, e, b)? No, missing edge. (e, d, b, d, e)? Yes. (f, c, f)? Yes.

Edge and vertex sets of paths

For a length-k path p, let $E(p) = \{\{p^i, p^{i+1}\} | i \in [1, k]\}$.

Let $V(p) = \{p^i | i \in [1, k+1]\}$. These are p's edge and vertex sets.

Find two cycles p and q such that $V(p) = V(q)$ but $E(p) \neq E(q)$.

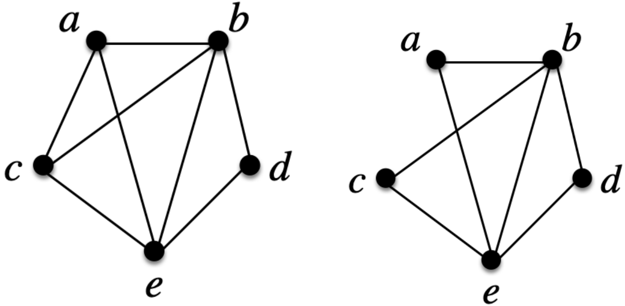


Tours on Graphs (on Tour)

Let G be a graph with n vertices and m edges. Let p be a length-k cycle in G.

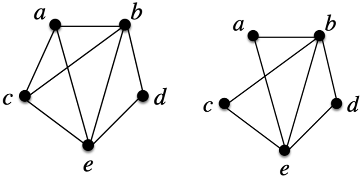
Definition: p is an *Eulerian tour* (edge tour) of G if $E(p) = E(G)$ and $k = m$.

Definition: p is a *Hamiltonian tour* (vertex tour) of G if $V(p) = V(G)$ and $k = n$.



Eulerian implies even

If a graph has an Eulerian tour, $\deg(v)$ must be even for all vertices.

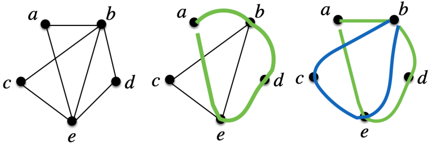


Proof sketch: Let p be an Eulerian tour of G. Every time the tour visits a vertex, it must arrive and depart (on fresh edges). Thus, each appearence of v on the cycle accounts for two edges that include v. In total, an even number.

Because it's an Eulerian cycle, every edge of the graph appears exactly once. Thus, every vertex of the graph is adjacent to an even number of other vertices.

Even implies Eulerian

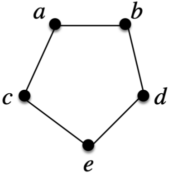
If all vertices in a connected graph have even degree, it has an Eulerian path (!).



Pick a vertex. Pick an arbitrary unused edge (if there is one) to go to another vertex. This process can only get stuck when we've returned to the starting vertex. That's the only vertex with an odd number of unused edges. If not an Eulerian tour, there's some vertex that we visited that has unused edges. That's because the graph is connected. Pick such a vertex and start this process again, rerouting our original route through this side route. By the earlier argument, the side route has to return to its starting point. Process must complete and can only complete with an Eulerian tour.

Cycle graph

If an n-node graph G has a cycle p that is both an Eulerian tour *and* a Hamiltonian tour, we call the graph a cycle and write it C_n.



Equivalent definition: C_n is connected and $\forall v \in V(G), \deg(v) = 2$.

Number of edges of C_n is n.

For what values of n, if any, does $K_n = C_n$?

Puzzles

- When does C_n have an Eulerian/Hamiltonian tour?
- When does K_n have an Eulerian/Hamiltonian tour?
- How many simple paths in K_n of length $n-1$?
- How many simple paths in C_n of length $n-1$?